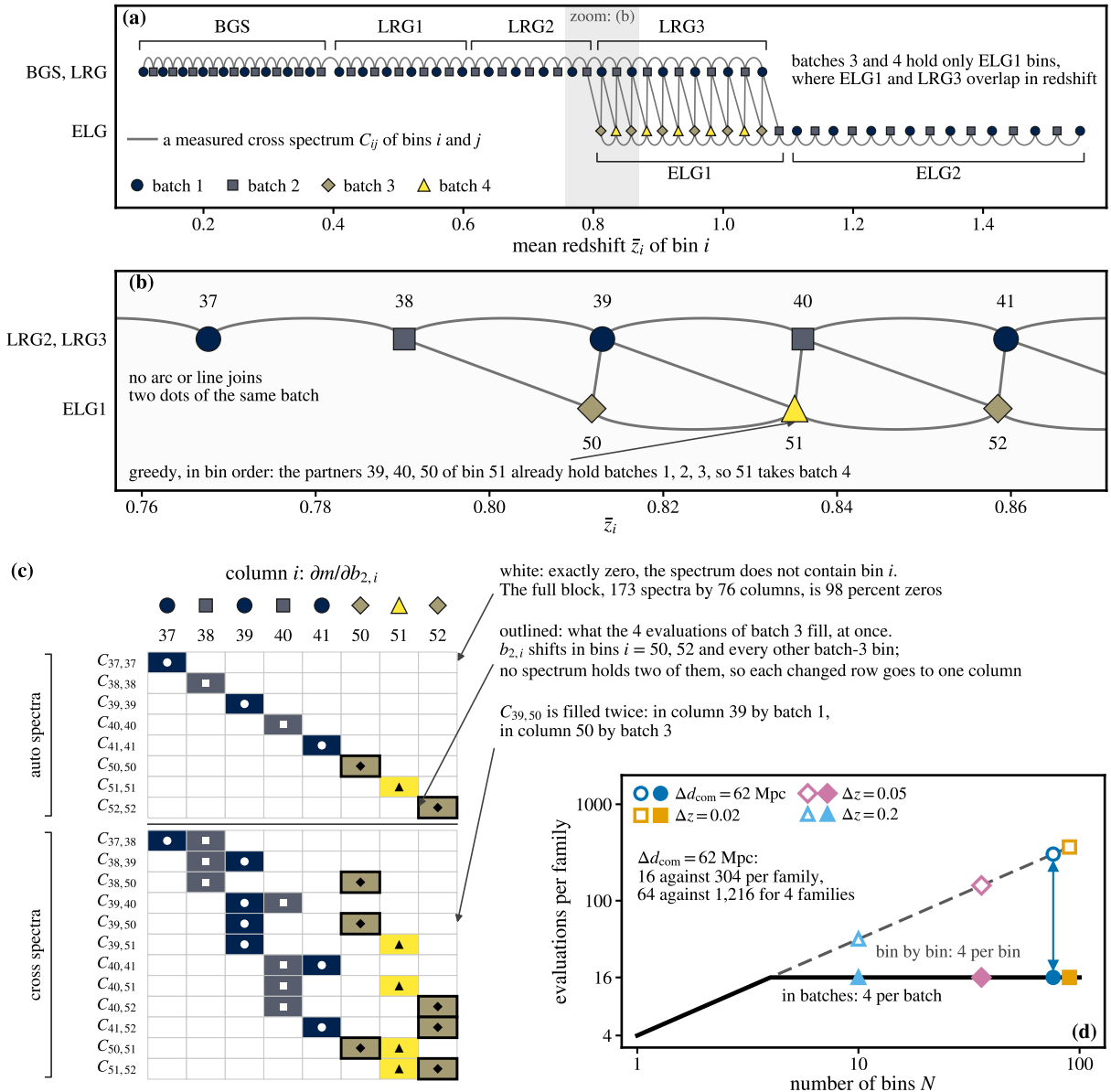

(a)
BGS
LRG1
LRG2
zoom: (b)
LRG3
BGS, LRG
ELG
a measured cross spectrum C_{ij} of bins i and j
batch 1
batch 2
batch 3
batch 4
batches 3 and 4 hold only ELG1 bins,
where ELG1 and LRG3 overlap in redshift
ELG1
ELG2
0.2
0.4
0.6
0.8
1.0
1.2
1.4
mean redshift $\bar{z}_i$ of bin i
(b)
37
38
39
40
41
LRG2, LRG3
ELG1
no arc or line joins
two dots of the same batch
50
51
52
greedy, in bin order: the partners 39, 40, 50 of bin 51 already hold batches 1, 2, 3, so 51 takes batch 4
0.76
0.78
0.80
0.82
0.84
0.86
$\bar{z}_i$
(c)
column i: $\partial m/\partial b_{2,i}$
37
38
39
40
41
50
51
52
auto spectra
$C_{37,37}$
$C_{38,38}$
$C_{39,39}$
$C_{40,40}$
$C_{41,41}$
$C_{50,50}$
$C_{51,51}$
$C_{52,52}$
cross spectra
$C_{37,38}$
$C_{38,39}$
$C_{38,50}$
$C_{39,40}$
$C_{39,50}$
$C_{39,51}$
$C_{40,41}$
$C_{40,51}$
$C_{40,52}$
$C_{41,52}$
$C_{50,51}$
$C_{51,52}$
white: exactly zero, the spectrum does not contain bin i.
The full block, 173 spectra by 76 columns, is 98 percent zeros
outlined: what the 4 evaluations of batch 3 fill, at once.
$b_{2,i}$ shifts in bins $i = 50, 52$ and every other batch-3 bin;
no spectrum holds two of them, so each changed row goes to one column
$C_{39,50}$ is filled twice: in column 39 by batch 1,
in column 50 by batch 3
$\Delta d_{\rm com} = 62$ Mpc
$\Delta z = 0.05$
$\Delta z = 0.02$
$\Delta z = 0.2$
$\Delta d_{\rm com} = 62$ Mpc:
16 against 304 per family,
64 against 1,216 for 4 families
bin by bin: 4 per bin
in batches: 4 per batch
1000
100
16
4
evaluations per family
1
10
100
number of bins N
(d)